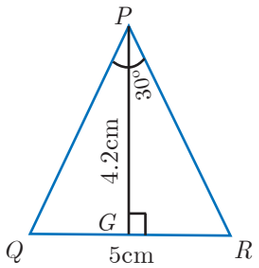
Construct a triangle when its base, the vertical angle and the altitude from the vertex to the base are given.

Example 4.18 Construct a triangle ΔPQR such that $QR = 5$ cm, $\angle P = 30°$ and the altitude from P to QR is of length 4.2 cm.



Rough diagram

Solution

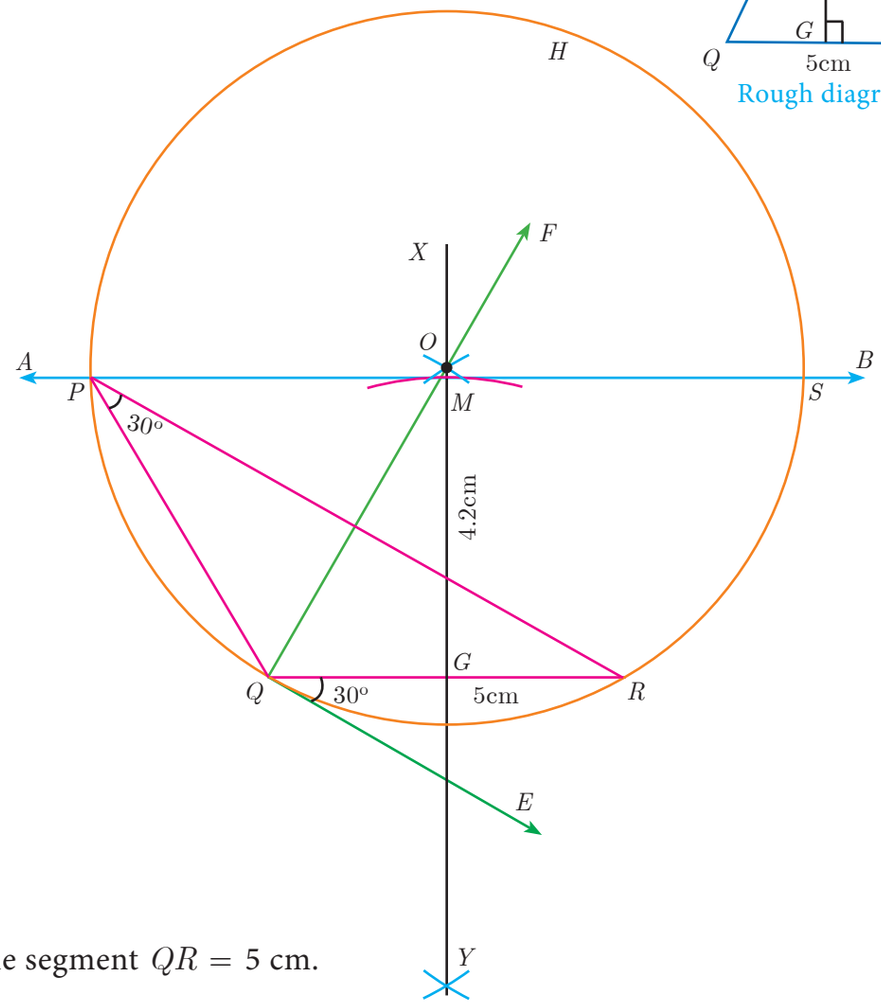


Fig. 4.43

Construction

Step 1 : Draw a line segment $QR = 5$ cm.

Step 2 : At Q draw QE such that $\angle RQE = 30°$.

Step 3 : At Q draw QF such that $\angle EQF = 90°$.

Step 4 : Draw the perpendicular bisector XY to QR which intersects QF at O and QR at G.

Step 5 : With O as centre and OQ as radius draw a circle.

Step 6: From G mark an arc in the line XY at M, such that $GM = 4.2$ cm.

Step 7 : Draw AB through M which is parallel to QR.

Step 8 : AB meets the circle at P and S.

Step 9 : Join QP and RP. Then $\triangle PQR$ is the required triangle.

Note

ΔSQR is another required triangle for the given measurements.

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